

Algebra 1

Unit 12

Lesson 5 Homework

Name Answer Key

Date _____

Period _____

A study shows that the number of times a student skips breakfast has a negative correlation with student test scores.

1. What does it mean that there is a negative correlation?

Students with higher numbers of skipped breakfasts have lower test scores.

2. Does this mean that not eating breakfast causes students to do poorly on tests?

No

3. What other factors might contribute to this association?

-Running late/rushing and not properly focused

-Students may not be hungry in the morning and also struggle with test material

4. What other information would be helpful to determine whether not eating breakfast causes lower test scores?

-Are the tests only offered in the morning?

At a theme park, there is a positive correlation between the amount of rainfall and the number of ponchos sold.

5. What does it mean that there is a positive correlation?

The days with higher rainfall have more ponchos sold.

6. Does this mean that increased poncho sales leads to more rainfall? If not, what does it mean?

No. Some may have purchased ponchos because rain was forecasted.

7. What other factors might contribute to this association?

Some may have bought ponchos to protect them on the water rides.

At a ski resort, as snow accumulation increases, so does the injury rate.

8. What kind of correlation does this association represent?

Positive correlation

9. Is there evidence of a causal relationship?

yes

10. What other information would be helpful to evaluate the claim of a causal relationship?

What are the injuries and how are they specifically related to snowfall?